

Bash Navigation

Cursor Movement

Shortcut	Description
Ctrl + A	Move cursor to the beginning of the line.
Ctrl + E	Move cursor to the end of the line.
Alt + B or Esc, B	Move backward one word.
Alt + F or Esc, F	Move forward one word.
Ctrl + B	Move backward one character.
Ctrl + F	Move forward one character.

Note: **Alt + ←** and **Alt + →** may work in some terminals, but they are **terminal-dependent** and are **not standard** Readline shortcuts.

- **Ctrl** shortcuts generally operate on **characters or lines**.
- **Alt (Meta)** shortcuts generally operate on **words**.
- **Esc** can be used instead of **Alt** by pressing **Esc**, releasing it, then pressing the next key (e.g., **ESC**, then **B**).

Editing Text

Shortcut	Description
Ctrl + K	Cut from the cursor to the end of the line.
Ctrl + U	Cut from the cursor to the beginning of the line.

Deleting Words

Shortcut	Description
Ctrl + W	Delete (cut) the word before the cursor.
Alt + Backspace	Delete the word before the cursor.
Alt + D or Esc, D	Delete the word after the cursor.

Deleting Characters

Shortcut	Description
Ctrl + H	Delete the character before the cursor (Backspace).
Ctrl + D	Delete the character under the cursor (Delete).

Copy/Paste, Undo, and Screen

Shortcut	Description
Ctrl + Y	Paste (yank) the last killed text.
Ctrl + _	Undo the last editing change.
Alt + R or Esc, R	Revert the current line to its original state from history.
Ctrl + L	Clear the screen (same effect as the <code>clear</code> command).

Command History Expansion

Command	Description
!!	Repeat the previous command.
!n	Run command number <i>n</i> from history.
!-n	Run the command that is <i>n</i> commands before the current one.
!string	Run the most recent command that starts with <code>string</code> .
!string:p	Print the matching command without executing it.

Command	Description
! \$	Expand to the last argument of the previous command.
* ! **	Expand to all arguments of the previous command (excluding the command name).

Command History Navigation

Shortcut	Description
Ctrl + R	Search command history incrementally (reverse search).
Ctrl + P	Previous command in history.
Ctrl + N	Next command in history.
Up Arrow	Previous command.
Down Arrow	Next command.
Alt + .	Insert the last argument of the previous command. Press repeatedly to cycle through earlier commands' last arguments.

Ctrl + R (Reverse History Search)

- Press **Ctrl + R**
- Type **dock**
- Shows the latest matching command.
- Press **Ctrl + R** again to cycle backward through older matches.
- Press **Enter** to use the selected command.

Example:

```
History:
  docker ps
  docker images
  docker compose up
  docker logs app
```

```
Ctrl + R → dock
```

1. docker logs app
2. docker compose up
3. docker images
4. docker ps

Shortcut: Ctrl + R = Search command history backward.

Bash Special Parameters

Parameter	Description	Example
\$0	Name of the script or shell	<code>./script.sh</code>
\$1	First argument	<code>hello</code>
\$2	Second argument	<code>world</code>
\${10}	Tenth argument (must use braces for 10+)	<code>\${10}</code>
\$#	Number of positional arguments	<code>3</code>
\$*	All positional arguments as a single word	<code>"a b c"</code>
@	All positional arguments as separate words	<code>"a" "b" "c"</code>
\$\$	PID (Process ID) of the current shell	<code>23541</code>
\$?	Exit status of the last command	<code>0, 1, 127</code>
\$_	PID of the last background process	<code>34121</code>
\$_	Last argument of the previous command	<code>file.txt</code>
\$-	Current shell option flags	<code>himBH</code>

Example

```
#!/bin/bash
```

```
echo "Script: $0"
```

```
echo "First: $1"
echo "Second: $2"
echo "Count: $#"
```

```
echo "All (*): $*"
echo "All (@): $@"
echo "PID: $$"
```

History Expansion (!)

Expansion	Meaning
!!	Previous command
!n	Run history entry number n
!-n	Run command n commands ago
!string	Last command starting with string
!?string?	Last command containing string
!\$	Last argument of previous command
!*	All arguments of previous command
^old^new	Repeat last command replacing first occurrence of old with new
!string:p	Print matching command without executing
!!:p	Print previous command
history	Show command history

Argument Designators (:)

These work after a history expansion.

Expansion	Meaning
!!:0	Command name
!!:1	First argument
!!:2	Second argument
!:\$	Last argument
!:*	All arguments
!!:^	First argument
!!:2-4	Arguments 2 through 4
!!:2-\$	Argument 2 to the last
!!: -1	All arguments except the last

Example

```
$ cp file1 file2 backup/
$ echo !:$
```

Expands to

```
echo backup/
```

Bash provides several convenient shortcuts:

Shortcut	Equivalent
!!	! -1 (previous command)
!^	!!:^ (first argument of previous command)
!\$	!:\$ (last argument of previous command)
!*	!:* (all arguments of previous command)
!:0	!!:0 (command name of previous command)
!:2	!!:2 (second argument of previous command)

When you omit the event (! !), Bash assumes **the previous command**.

So these are identical:

```
echo !$  
echo !!:$  
echo !-1:$
```

`^old^new` Repeat last command replacing first occurrence of `old` with `new`

Example: 1

```
$ cat myfiel.txt  
cat: myfiel.txt: No such file or directory
```

Correct it:

```
$ ^myfiel^myfile
```

Bash executes:

```
cat myfile.txt
```

Example: 2

```
grep "error" logfile.log  
grep: logfile.log: No such file or directory
```

```
$ ^logfile^server
```

This reruns:

```
grep "error" server.log
```